

Clinical Neuroscience

NSC 4319 Section 2

Spring 2026

3 credits, lecture-based course

Time: 10:10 AM - 11:00 AM M/W/F

Location: BSB A235

Instructor Information

Instructor: Dr. Nicole Phillips, Lecturer

E-mail: Nicole_A_Phillips@baylor.edu

Office: BSB A212

Office hours: Tuesdays 2 – 3 PM and Wednesdays 12:15 – 1:15 PM. I am usually free most of the day on Tuesday and Thursday, but please email me to make sure I am available at a time that will work for you.

This syllabus represents an agreement between the professor and the student regarding the nature and guidelines of this course. The professor reserves the right to modify, if necessary, the syllabus to adapt to new circumstances. These changes will be made known to students via Canvas, email, and/or in class.

Required Text and Materials

Textbook: Augustine, G. J., Groh, J. M., Huettel, S. A., LaMantia, A.-S., White, L. E., & Purves, D. (2023). *Neuroscience* (7th ed.). New York: Oxford University Press.

This text is sold at the Baylor bookstore as well as other online retailers (Amazon, Barnes & Noble, Textbooks.com, etc...)

Students may purchase or rent whatever version of the text fits their preference or budget: (1) Paperback, (2) Loose-leaf, (3) eBook.

Any additional readings will be posted as PDF documents available on Canvas

Course Overview

Course Description: The purpose of this course is to help you establish a foundational understanding of how the nervous system functions so that you can better understand the mechanisms underlying neurological and neuropsychiatric disorders. To achieve this goal, we will discuss the molecular and cellular underpinnings of nerve conduction, synaptic transmission, neural plasticity, neuroanatomy, and neural circuits. We will then consider how these principals underlie processes, such as somatosensation,

movement, etc... We will also consider how dysfunction in these processes can lead to conditions such as pathological pain, mood disorders, developmental disorders, and neurodegenerative diseases. I aim to cultivate an environment of curiosity, respect, and compassion as we seek to better understand the complexities of clinical neuroscience.

The lab is an additional course for which you will register separately. Your lab grade has no impact on your grade in this course. Also, your grade in this course has no impact on your final lab grade. However, be advised that **taking the lab concurrently with the lecture course can have tremendous academic advantage.** There are topics that are covered in lecture that are put into practical terms in the lab.

Course Objectives:

By the time you complete this course you will be able to:

1. **Explain** basic principles underlying cellular communication within the nervous system.
2. **Describe** how these processes allow for learning, somatosensation, and movement.
3. **Identify** symptoms of various neurological and neuropsychiatric disorders.
4. **Understand** how dysfunction in neuronal communication relates to various neurological and neuropsychiatric disorders.

If at any time you feel the class is straying from these objectives please let me know and I will adjust the course if necessary

Grading and Assignments

Grading Scale: Assignments and final grades will be based on the following scale:

Percentage	Letter Grade
≥ 90.0%	A
88.00% - 89.99%	B+
80.00% - 87.99%	B
78.00% - 79.99%	C+
70% - 77.99%	C
60% - 69.99%	D
<60%	F

Note that percentage ranges are exact. I do not round final grades.

Assignments: Final grades will be calculated based on the following assignments:

3 Non-cumulative Tests	60%
Cumulative Final Test	20%
Disorder Presentation	10%
Concept Checks	5%
Case Vignettes	5%

Non-cumulative Tests (60%): There will be three non-cumulative tests throughout the semester (worth 20% of your final grade each). The questions will be based on the material covered in the preceding lectures and assigned textbook readings. Question format will include multiple choice and short answer. You will have the full class period to complete each test. All tests must be taken during the scheduled class period unless the student has documented accommodations with the Office of Access and Learning Accommodation.

Cumulative Final Test (20%): The final test will be cumulative and include information that was covered in the text as well as in class and on all three of the previous tests. The format of the test will be the same as the non-cumulative tests, just double the number of questions. The final test for this course will be on **Monday 5/11/26 from 2:00 – 4:00 PM.**

Disorder Presentations (10%): There is not enough time in the semester to cover every neurological or neuropsychiatric disorder. Therefore, your assignment is to choose a neurological or neuropsychiatric disorder **not** discussed in class and assemble one PowerPoint slide about the disorder. Your slide should include a short description of the disorder, symptoms associated with the disorder, and what we currently know about dysfunction of the nervous system as it relates to symptomology. You will also include one additional slide (2 total slides) with APA formatted references. See Canvas for a grading rubric, an example, and the list of disorders we will cover in class. Presentations will take place on **Monday 5/4**. This assignment will help expose you and your classmates to additional neurological conditions and also help you practice your presenting skills in a supportive environment.

Concept Checks (5%): Low-stakes multiple choice quizzes (referred to as concept checks) will be due at the end of each week when there is not a test. Concept checks will be completed through Canvas. You will have 15 minutes to answer 10 questions. Each quiz must be completed by midnight on Sunday, before the next class week begins (see course schedule). You will be given two attempts for each quiz. If you are happy with your score from your first attempt, you do not need to do the other attempt. If

you do complete another attempt, the higher score of the attempts will be recorded. Be aware that because the quiz utilizes a test bank, you may be presented with new questions on each attempt. I encourage you to complete each concept check without the use of outside resources (e.g. notes, textbook, study materials), as this will help you prepare for the environmental conditions of the tests. The concept checks will also help you to utilize the testing effect to improve your retention of learned material (Karpicke, 2009).

Case Vignettes (5%): For these assignments, you will **first** write your own case vignette about a patient with a particular disorder (e.g. major depressive disorder). Your vignette should be about a paragraph in length and include relevant factors for diagnosis of the particular disorder (e.g. gender, age, motor symptoms, sensory symptoms, cognitive symptoms, elements of dysfunction and/or distress etc...). Once you have written your own case vignette, you will use an artificial intelligence (AI) of your choosing (e.g. ChatGPT, Claude, Copilot etc...) to write another case vignette for the same disorder. Once the AI has generated the vignette, your task is to compare and contrast the AI generated vignette with your own. **You are not allowed to use AI to write the comparison.** In your comparison, you will answer the following questions.

1. What is similar between your vignette and the AI vignette? What is different?
2. Are there any parts of the AI vignette that are inaccurate or highly unlikely for the given diagnosis?
3. Is there anything missing from the AI vignette that you think would be important to add?

The comparison narrative should be at least a paragraph in length (double spaced, Arial 12pt font). The document you submit on Canvas should include your vignette, the AI generated vignette, and your comparison narrative. Case vignettes will also be due by the end of the day the Sunday after the topic is covered in class.

The case vignette assignments are meant to help you practice thinking critically about neurological disorders/diseases and how symptoms may manifest.

Canvas

All class materials will be posted on the class Canvas website. Materials will include a copy of the syllabus, grades, answer keys for worksheets, and other helpful documents. Any announcements regarding the course or course material will also be posted on Canvas.

The grades shown on Canvas will serve only as an **estimate** of your final grade in the class. If students are curious about their progress in the class, they should contact the course instructor to get a more accurate estimate of their grade.

Course Policies:

Attendance: To earn course credit in the College of Arts and Sciences, a student must attend at least 75% of all scheduled class meetings. Attendance will be taken every day

at the beginning of class. Any student who does not meet this minimal standard will automatically receive a grade of “F” in the course.

Asynchronous Instruction Day: The University Calendar includes an asynchronous day of instruction for M/W/F classes during the spring semester. The asynchronous day for this class will be **Sunday 4/19** and will serve as a workshop day for your disorder presentation. A draft of your presentation must be submitted by the end of the day.

Test Preparation and Conduct: Tests will be completed in-person on paper. Bring a pencil to each test. Make sure to legibly write your name, student ID, section number (5), and test version (A or B) on both your test booklet and scantron. Students must turn off all electronic devices (e.g., cell phones, computers, etc.) and place them out of sight in a closed backpack/purse before the test starts. Students are only permitted pencils/pens, erasers, one beverage, and an analog watch (no cell phone, smart watch, headphones, airpods, etc. allowed) while taking the test. Students will not be allowed to take bathroom breaks unless a previous accommodation is in place or unless previously discussed with the instructor, so plan ahead. Students may not wear a hat or hoodie during a test. Students should not engage in suspicious behaviors (e.g., glancing at other students’ tests, tapping fingers/feet, having wandering eyes, talking to others, or wearing headphones and sunglasses). Violating these policies is cause for dismissal from the test. Any possible cheating will result in a **0** grade and generation of a report with the honor council. If you arrive late for a test and a student has already completed the test and left the classroom, the test will be considered “missed” and the Make-up test policy will apply.

Late work: All assignments are due by the specified due date on the syllabus (see course schedule below). I will not accept an assignment that is turned in late unless there are documented extenuating circumstances. Extenuating circumstances are any events outside of the student’s control that may prevent them from being in class/turning in work (e.g. family member passed away, car accident, etc...). Documentation must be provided in one of the following forms: (1) a note from a physician explicitly stating that the student could not be in class/turn in the assignment; (2) notification from the Baylor University Chaplain’s Office that there has been a death or serious injury in the family; or (3) a note from a university official for a university-sponsored event. Late work without an extenuating circumstance approved by the instructor will be an automatic **zero**.

Make up tests: If an extenuating circumstance prevents you from taking a test, you may request one make-up opportunity and you will be asked to provide documentation for the test absence. If a situation arises, please email me as soon as possible. Please note, **choosing to purchase a plane ticket on a scheduled class day will not count as an extenuating circumstance**.

If you are approved to take a make-up test, you must take the make-up test within one week of the original test date. You must coordinate your make-up test with the instructor. All make-up tests will be in the same format as the other tests, but they will have different questions.

Cancellation Policy: If an unexpected circumstance arises during the semester and class needs to be canceled, I will post an announcement on Canvas as soon as possible.

Student Athletes: If you are on an athletic team and you are travelling on any class or test days, please provide me with a letter listing the dates you will miss as soon as possible. This list must be signed off by your coach or other authorized team representative. It is **up to you** to find out now whether there will be a team-related conflict and communicate that to me. The final test will not be rescheduled due to team-related conflicts.

Academic Integrity: Plagiarism or any form of cheating involves a breach of student-teacher trust. This means that any work you submit is expected to be your own, neither composed by anyone else as a whole or in part, nor handed over to another person for complete or partial revision. Be sure to document all ideas that are not your own. The rules regarding academic integrity are detailed in the Baylor University Honor Code: www.baylor.edu/honorcode (Links to an external site.). It is your responsibility to be familiar with and uphold these rules. If you become aware of another student who violates these rules, you are encouraged to report the incident to your instructor. These reports will be confidential. If you are acting in a way that suggests you may be engaged in academic misconduct, I am obligated by college policy to take immediate action to protect the integrity of the course. Any student who violates these rules will automatically be given a **zero** and the event will be reported to the Honor Council, **which may result in a grade of “F” for the entire course.**

Use of Artificial Intelligence: All assignments should be your original work and should not be produced in part or in total with the assistance of artificial intelligence (for example, ChatGPT, Grammarly, or some other resource). Use of artificial intelligence without my explicit permission constitutes a violation of the Honor Code at Baylor University. The only exception to this policy is the case vignette assignment.

Cell Phones and other electronic devices: During class time, all cell phones should be turned off and put away, unless the instructor grants permission to use cell phones for an in-class activity. If you need to leave a phone on for emergency purposes, please let the instructor know before class begins. You may use a Laptop/iPad/e-Notebook for taking notes but please keep in mind that web surfing is distracting to other students and it robs you of engaging with the class. If the instructor catches you using an electronic device for anything other than taking notes (e.g. web surfing, reading Twitter feeds, updating Facebook status etc.), your privilege to use electronic devices for notetaking will be revoked for the rest of the semester and you will need to take notes by hand.

Course etiquette: All students in this class are adults (or a few months away from being adults), so I will expect you to behave as such, as that is how I will treat you. To ensure a climate of learning for all, disruptive or inappropriate behavior (repeated

outbursts, disrespect for ideas of others, etc...) may result in removal from this class. This class will involve many situations where students are required to share their personal thoughts and ideas. I want students to be able to do this without being afraid of ridicule or personal attack. In short, treat others how you would like to be treated.

Email etiquette: All correspondence outside of class will be done through Canvas or email. Emails should be professional and appropriate and include a proper subject (NSC 4319 *issue*), greeting (Hi Professor Phillips, or Dear Dr. P), and signature. The body of the email should be a concise description of your question. Please check the syllabus before emailing me. I will respond to all emails within 1 business day during the week. I will respond to emails received after 6 PM the following day, so if a question is important, please plan accordingly. Emails received during the weekend may not be answered until the following business day.

Academic Accommodations: Any student who needs learning accommodations should inform the instructor immediately at the beginning of the semester. **The student is responsible** for obtaining appropriate documentation and information regarding needed accommodations from Baylor University's Office of Access and Learning Accommodation (<https://oala.web.baylor.edu/>) and providing it to the instructor early in the semester. OALA is part of the Paul L. Foster Success Center located in Sid Richardson building room 190 and can be contacted at 254-710-3605.

University Resources

Your success in this course is important to me and I encourage you to reach out to me if you find that you are struggling with course materials. Baylor also offers numerous academic and career resources that may be helpful to you both in this class and during your time here at Baylor.

Office of Access and Learning Accommodation	(254) 710-3605
Academic Support Services	(254)-710-8696
First in Line Program	(254)-710-6854
VETS Program	(254)-710-7264
Information Technology Help Desk	(254)-710-4357
Counseling Center	(254)-710-2467
University Writing Center	(254) 710-4849

Title IX Statement: Baylor University does not discriminate on the basis of sex or gender in any of its education or employment programs and activities, and it does not tolerate discrimination or harassment on the basis of sex or gender. If you or someone you know would like help related to an experience of sexual violence including sexual assault, harassment, domestic violence, dating violence, stalking or other type of non-consensual sexual conduct, please contact the Title IX Office at (254-710-8454).

Anonymous reporting for students or third parties is also available on the Title IX website at <https://www.baylor.edu/TitleIX/> (Links to an external site.).

The Title IX office understands the sensitive nature of these situations and can provide information about available on- and off-campus resources, such as counseling and psychological services, medical treatment, academic support, university housing and other forms of assistance. Staff members at the office will also explain your rights and the judicial process options, if you choose to file a complaint with the University. You will not be required to share your experience, and the Title IX Office will keep any information private. The Title IX Office exists to support and empower students, while allowing them to remain in control. **If you or someone you know feels unsafe or may be in imminent danger, please call the Baylor Police Department (254-710-2222) or Waco Police Department (9-1-1) immediately.** For more information on the Title IX Office, the Sexual and Gender-Based Harassment and Interpersonal Violence policy, reporting, and resources available, please visit the website provided above.

Course Schedule

Week	Dates	Topic	Readings & Assignments
1			
	W 1/21	Syllabus Overview	
	F 1/23	Studying the Nervous System	Ch. 1
2	M 1/26	Studying the Nervous System	
	W 1/28	Electrical Signals of Nerve Cells	Ch. 2
	F 1/30	Electrical Signals of Nerve Cells	Concept Check #1
3	M 2/2	Voltage-dependent membrane permeability	Ch. 3
	W 2/4	Voltage-dependent membrane permeability	Multiple Sclerosis Case Vignette
	F 2/6	Synaptic Transmission	Ch. 5 Concept Check #2
4	M 2/9	Synaptic Transmission	
	W 2/11	Review for Test #1	Selection of disorder presentation topic
	F 2/13	Test #1 (Ch. 1-3 & 5)	
5	M 2/16	Neurotransmitters and their Receptors	Ch. 6
	W 2/18	Neurotransmitters and their Receptors	
	F 2/20	Molecular Signaling Within Neurons	Ch. 7 Concept Check #3

6	M 2/23	Molecular Signaling Within Neurons	
	W 2/25	Synaptic Plasticity	Ch. 8
	F 2/27	Synaptic Plasticity	Autism Spectrum Disorder Case Vignette Concept Check #4
7	M 3/2	Touch and Proprioception	Ch. 12
	W 3/4	Touch and Proprioception	
	F 3/6	Pain and Temperature	Ch. 13 Concept Check #5
8	M 3/9	No Class Spring Break	
	W 3/11		
	F 3/13		
9	M 3/16	Pain and Temperature	
	W 3/18	Review for Test #2	
	F 3/20	Test #2 (Ch. 6-8 & 12-13)	
10	M 3/23	Lower motor neuron circuits and motor control	Ch. 16
	W 3/25	Lower motor neuron circuits and motor control	
	F 3/27	Upper motor neuron control of the brainstem and spinal cord	Ch. 17 List of 5 references for disorder presentation Concept Check #6
11	M 3/30	Upper motor neuron control of the brainstem and spinal cord	
	W 4/1	Modulation of Movement by the Basal Ganglia	Ch. 18 Parkinson's Disease Case Vignette
	F 4/3	No Class Easter Break	
12	M 4/6		
	W 4/8	Modulation of Movement by the Basal Ganglia	
	F 4/10	Modulation of Movement by the Cerebellum	Ch. 19 Concept Check #7
13	M 4/13	Modulation of Movement by the Cerebellum	
	W 4/15	Review for Test #3	

	F 4/17	Test #3 (Ch. 16-19)	
14	Su 4/19	Asynchronous Presentation Workshop	Draft of Presentation
	M 4/20	Cognitive Functions and Organization of the Cerebral Cortex	Ch. 27
	W 4/22	Cognitive Functions and Organization of the Cerebral Cortex	
	F 4/24	Cortical States	Ch. 28 Concept Check #8
15	M 4/27	Cortical States	
	W 4/29	Emotion	Ch. 32
	F 5/1	Emotion	Major Depressive Disorder Case Vignette Concept check #9
16	M 5/4	Disorder Presentations	
	W 5/6	Review for Final Test	
Cumulative Final Test Monday 5/11/26, 2:00 PM – 4:00 PM			

References

Karpicke, & Jeffrey, D., (2009). Metacognitive control and strategy selection: Deciding to practice retrieval during learning. *Experimental Psychology*. 138(4) 469-486